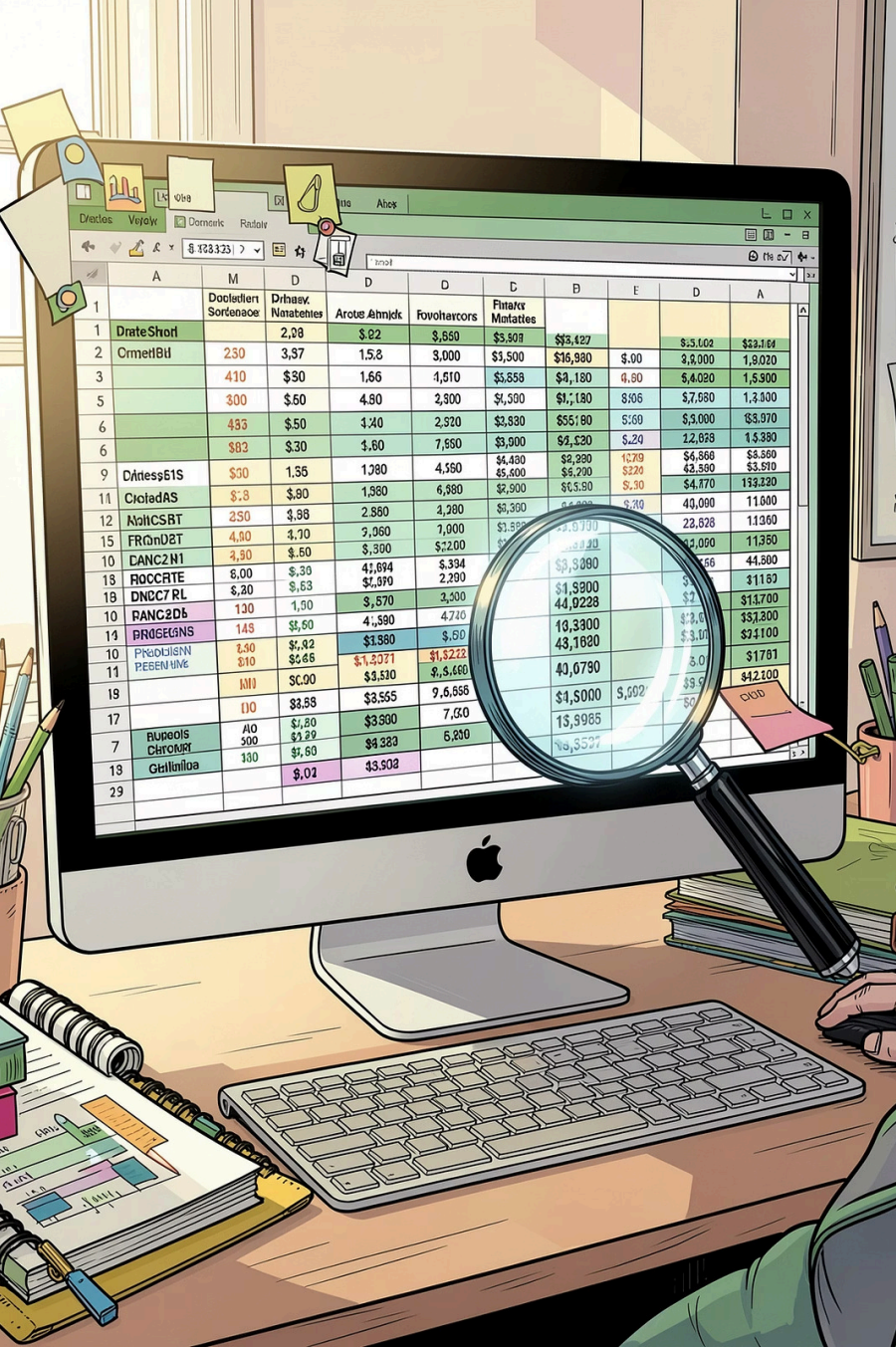




Customer Shopping Behavior Analysis

Uncovering insights from 3,900 transactions across product categories to guide strategic decisions.



Dataset Summary

3,900 Rows

18 columns of
transactional data

Key Features

Demographics,
purchase details,
shopping
behavior

Missing Data

37 null values in
Review Rating

Exploratory Data Analysis in Python



01

Data Loading

Imported dataset using pandas

02

Cleaning

Imputed missing ratings by category median

03

Feature Engineering

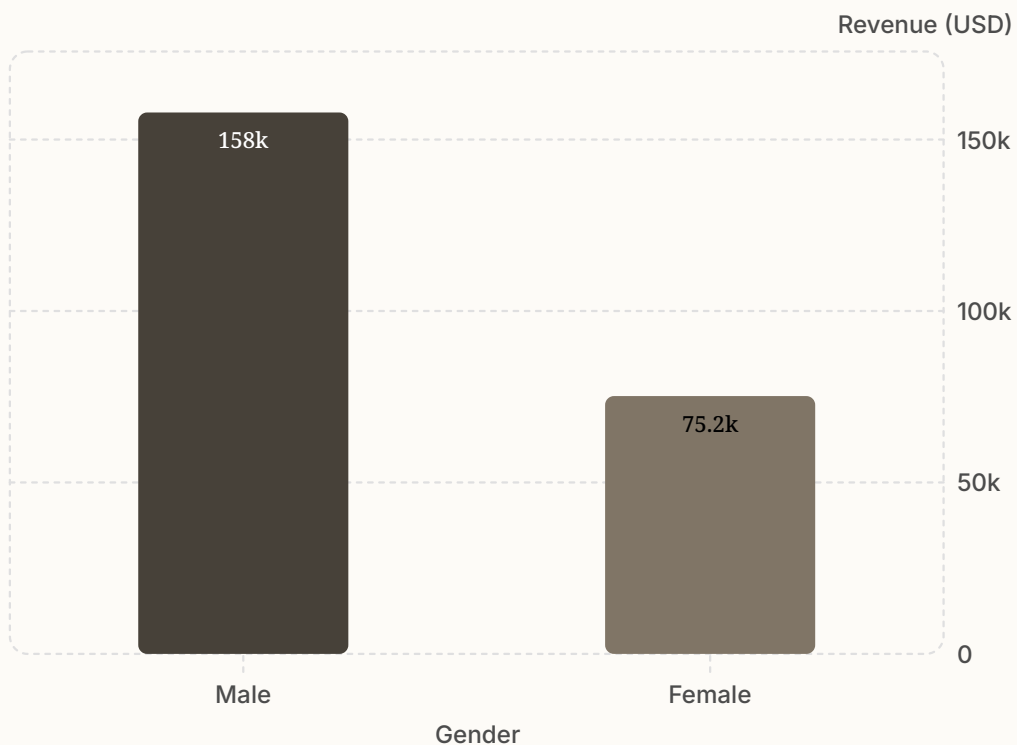
Created age_group and purchase_frequency_days

04

Database Integration

Loaded cleaned data into PostgreSQL

Revenue & Spending Insights



Key Findings

Male customers generated **2.1x** more revenue than female customers.

High-spending discount users identified — many spent well above the \$59.76 average purchase amount.

Top Products by Rating



Gloves — 3.86



Sandals —
3.84



Boots — 3.82

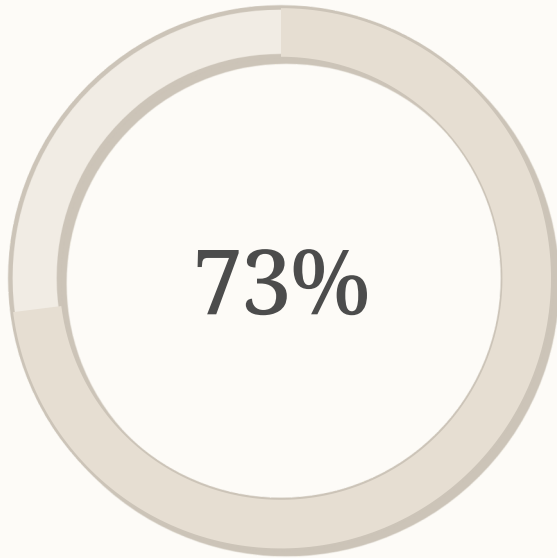


Hat — 3.80

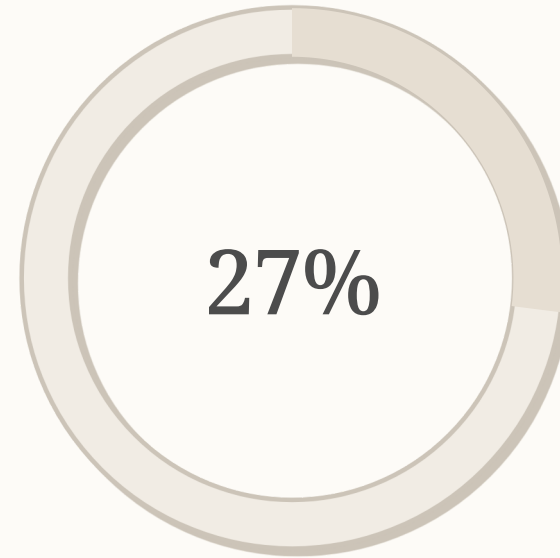


Skirt — 3.78

Subscribers vs. Non-Subscribers



Non-subscribers



Subscribers

Non-subscribers drove **\$170,436** in total revenue vs. **\$62,645** from subscribers.

Average spend is nearly identical: \$59.87 vs. \$59.49.

Express shipping averages **\$60.48** vs. Standard at \$58.46.

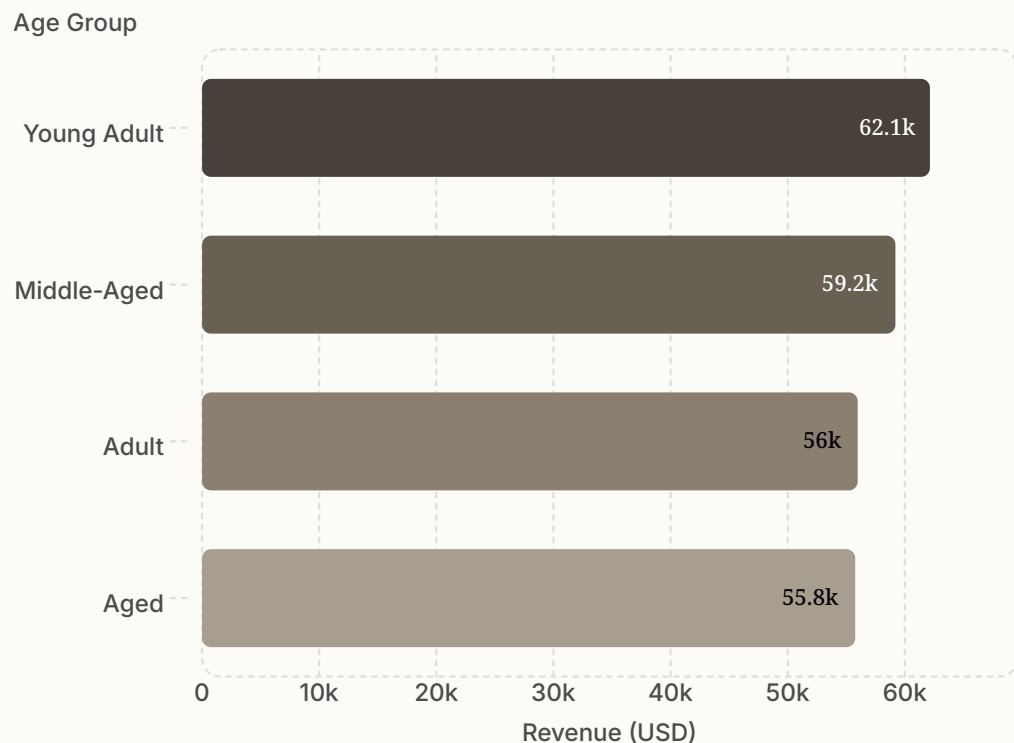
Customer Segmentation



The vast majority of customers are **Loyal** (3,116), with 701 Returning and just 83 New.

Repeat buyers with >5 purchases: **958** subscribed, **2,518** did not.

Revenue by Age Group



Young adults lead revenue at **\$62,143**, closely followed by middle-aged at \$59,197.

Top discount-dependent products: Hat (50%), Sneakers (49.7%), Coat (49.1%).

Power BI Dashboard

Interactive visualizations tracking 3.9K customers, \$59.76 avg purchase, 3.75 avg rating, and revenue across categories and age groups.



Business Recommendations

Focus efforts on high-revenue segments and express-shipping users to maximize impact.

